

Padé oscillators and a new formulation of perturbation theory

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Being guided by the problem of bound states in potentials close to their Padé approximants, a new Rayleigh–Schrödinger-type perturbation theory is developed. The unperturbed system is understood here in a broader sense: its solutions are not needed, but merely the related nondiagonal unperturbed propagator R . In particular, all the chain models $H_0\psi = ES_0\psi$ (H_0, S_0 = band matrices) with arbitrary perturbations are then perturbatively solvable, with R constructed in terms of auxiliary matrix continued fractions f_n . Alternatively, a “generalized unperturbed spectrum” $\hat{f}_n$ may be required as an input: The algebraically constructed asymptotics of the f_n ’s play this role in our Padé examples. Due to $S \neq I$, the “Sturmians” may also be constructed. In the test evaluations of the binding energies and/or couplings, the simultaneous upper and lower bounds of high precision are shown to be numerically obtainable.

I. INTRODUCTION

Usually, the textbook¹ Rayleigh–Schrödinger (RS) perturbation theory is applied to the Hamiltonian matrices that are almost diagonal,

$$\langle m|H|n \rangle = \begin{cases} \text{large,} & m = n, \\ \text{small,} & m \neq n. \end{cases} \quad (1.1)$$

Recently, a weaker assumption

$$\langle m|H|n \rangle = \begin{cases} \text{large,} & |m - n| \leq t, \\ \text{small,} & |m - n| > t, \quad t \geq 0, \end{cases} \quad (1.2)$$

has been shown to be tractable in the same spirit.² Here, we intend to describe a generalization of the latter modified RS (MRS) perturbation theory to an even more flexible class of Hamiltonians.

As an inspiration, let us recall the radial Schrödinger one-body equation

$$\left[-\frac{d^2}{dr^2} + \frac{l(l+1)}{r^2} + V(r) \right] \psi(r) = E\psi(r), \quad l = 0, 1, \dots, \quad (1.3)$$

with the often-studied potential³

$$V(r) = \frac{a + br^2 + cr^4}{1 + dr^2} \sim r^2 + \frac{\mu r^2}{1 + dr^2}, \quad c > 0, \quad d > 0, \quad (1.4)$$

all of the matrix elements of which are nonzero in the standard harmonic oscillator basis⁴ [i.e., $t \gg 1$ in (1.2)]. Nevertheless, an elementary modification of Eqs. (1.3) and (1.4),

$$[S(r)H^{(\text{oscillator})} + \mu r^2]\psi(r) = ES(r)\psi(r), \quad S(r) = 1 + dr^2 > 0, \quad (1.5)$$

acquires the infinite tridiagonal-matrix form

$$\sum_{n=m-1}^{m+1} [\langle m|(SH^{(\text{oscillator})} + \mu r^2)|n \rangle - E \langle m|S|n \rangle] \langle n|\psi \rangle = 0, \quad m = 0, 1, \dots, \quad (1.6)$$

in the same basis, with $S \neq I$ but, at the same time, with the extremely simple form of matrix elements (Whitehead *et al.*⁵).

Our present paper will be devoted to all the generalized eigenvalue problems⁶ of similar type:

$$(H - ES)|\psi \rangle = 0, \quad S \neq I. \quad (1.7)$$

For the sake of definiteness, we shall assume that

$$H = T + \lambda W, \quad S = S_0 + \lambda S_1, \quad (1.8)$$

where both T and S_0 are band matrices and $\lambda \ll 1$.

In the first, general part of our study (Sec. II), we shall introduce a rearrangement of Eq. (1.8),

$$H = H_0 + \lambda H_1, \quad H_0 = T + U, \quad \lambda H_1 = \lambda W - U, \quad (1.9)$$

where $U = O(\lambda)$ will be specified in a “self-consistent” way. Then we shall describe the related generalized RS (GRS) construction of expansions,

$$E = E_0 + \lambda E_1 + \lambda^2 E_2 + \dots, \quad (1.10)$$

$$|\psi \rangle = |\psi_0 \rangle + \lambda |\psi_1 \rangle + \lambda^2 |\psi_2 \rangle + \dots,$$

the complexity of which remains, roughly speaking, on the MRS level.

In Sec. III, an illustration of applicability of the overall GRS scheme may be found: We consider arbitrary $V(r)$ in Eq. (1.3), and an arbitrary Padé approximant⁷ of this function is incorporated into the band-matrix component T of the Hamiltonian. A $(2t + 1)$ -diagonal equation of the type (1.6) is obtained as a simple special case of Eq. (1.7).

In Sec. IV, a matrix continued fractional (MCF) construction of the propagator R is described, with a particular emphasis laid upon the MCF convergence for $V(r) = V_{\text{Padé}}(r)$. Nevertheless, only Sec. V makes our perturbative formalism complete: It resorts to the use of $\hat{f}_n$ ’s (the “capped” input MCF-like quantities) and simplifies the whole GRS construction considerably. Its final form may be characterized by the capped rearrangement of Eq. (1.9),

$$H = H_0 + \lambda H_1, \quad (1.11)$$

$$H_0 = T + U + \hat{\Theta} = \hat{T} + U,$$

$$\lambda H_1 = \lambda W - U - \hat{\Theta} = H - H_0,$$

$$U = O(\lambda), \quad \hat{\Theta} = \hat{T} - T = O(\lambda),$$

and makes the traditional RS unperturbed solutions redundant.

In the Padé-oscillator examples, the role of the $\hat{f}_n$ “capped input” is assigned to the so-called fixed point asymptotic MCF estimates. These quantities define the propagator $R = \hat{R}$ as well as the capped operators $\hat{T}$ and $\hat{Q}$ in Eq. (1.11). In a way, they are analogs to the ordinary RS unperturbed spectrum. An example of their self-consistent variation is studied numerically and shown to be able to provide the upper and lower bonds to the energies.

II. THE GRS PERTURBATION THEORY

An insertion of the RS-type ansatz (1.9) and (1.10) in the perturbed chain model (1.7) leads immediately to the RS hierarchy of requirements arranged as the separate $O(\lambda^k)$ identities. The first one is a $(2t + 1)$ -diagonal chain model⁸

$$\sum_{n=m-t}^{m+t} (\langle m|H_0|n\rangle - E_0\langle m|S_0|n\rangle) \langle n|\psi_0\rangle = 0, \quad m = 0, 1, \dots, \quad (2.1)$$

while the rest of them reads

$$(H_0 - E_0 S_0)|\psi_k\rangle + H_1|\psi_{k-1}\rangle = \sum_{m=1}^k (E_m S_0 + E_{m-1} S_1)|\psi_{k-m}\rangle, \quad k = 1, 2, \dots \quad (2.2)$$

A. $k=0$

In contrast to the standard RS theory that requires the operator $H_0 - E S_0$ diagonal, the $t \geq 1$ Eq. (2.1) is not solvable in general. Fortunately, we may circumvent this difficulty in full analogy with the $S = I$ MRS case,² i.e., via an introduction of an auxiliary separable field U ,

$$H_0 = T + U, \quad \lambda H_1 = \lambda W - U, \quad U = |0\rangle g \langle 0|. \quad (2.3)$$

With the sufficiently small coupling strength,

$$g = O(\lambda), \quad (2.4)$$

this split of the Hamiltonian remains still compatible with (1.10).

In the same MRS spirit, we may introduce also the projectors $P = |0\rangle \langle 0|$ and $Q = 1 - P$ and define the auxiliary unperturbed propagator as an inverse matrix,

$$R = Q(P - QD_0Q)^{-1}Q, \quad D_0 = T - E_0 S_0. \quad (2.5)$$

Our (assumed) knowledge of this operator enables us to eliminate the “irrelevant” components of the wave functions

$$Q|\psi_0\rangle = RD_0|0\rangle \langle 0|\psi_0\rangle, \quad \langle 0|\psi_0\rangle \neq 0 \quad (2.6)$$

from (2.1). This reduces our unperturbed $k=0$ Schrödinger equation to mere scalar relation

$$g = -\langle 0|D_0|0\rangle - \langle 0|D_0 R D_0|0\rangle. \quad (2.7)$$

Now a key point of the construction is a reinterpretation of this chain-model eigenvalue condition as a definition of $g = g(E_0)$. Then Eq. (2.1) becomes an identity, the zeroth-order quantities $\langle 0|\psi_0\rangle$ and E_0 remain free parameters, and only the condition of smallness (2.4) must be taken into account as a restriction of variability of the latter parameter.

B. $k \geq 1$

We may pick up the most natural normalization

$$\langle 0|\psi_0\rangle = 1, \quad \langle 0|\psi_k\rangle = 0, \quad k = 1, 2, \dots, \quad (2.8)$$

and rewrite the P projection of Eq. (2.2) as a definition of energies

$$\begin{aligned} E_k \langle 0|S_0|\psi_0\rangle &= \langle 0|D_0|\psi_k\rangle + \langle 0|W|\psi_{k-1}\rangle \\ &\quad - \delta_{k,1} \cdot h_0 - E_0 \langle 0|S_1|\psi_{k-1}\rangle \\ &\quad - \sum_{m=1}^{k-1} E_m \langle 0|S_0|\psi_{k-m}\rangle \\ &\quad + \langle 0|S_1|\psi_{k-m-1}\rangle, \quad k = 1, 2, \dots \end{aligned} \quad (2.9)$$

Here, the symbol δ_{ij} is a Kronecker delta ($\sim \langle 0|\psi_{k-1}\rangle$) and $g/\lambda = h_0 = O(1)$ [cf. Eq. (2.4)]. Similarly, the Q -projected remainder of Eq. (2.2) defines the wave functions

$$\begin{aligned} |\psi_k\rangle &= R W |\psi_{k-1}\rangle - \sum_{m=1}^k E_m R S_0 |\psi_{k-m}\rangle \\ &\quad - \sum_{m=1}^k E_{m-1} R S_1 |\psi_{k-m}\rangle, \quad k = 1, 2, \dots \end{aligned} \quad (2.10)$$

C. The iterative interpretation of formulas

In principle, the coupled pair of the $t \geq 1$ equations (2.9) and (2.10) defines the explicit RS-like multiple series representation of corrections E_k and $|\psi_k\rangle$ in (1.10). For example, we may eliminate, say, $|\psi_k\rangle$ from (2.9) and get

$$\begin{aligned} E_k &= \kappa (\langle \varphi_0|W|\psi_{k-1}\rangle - h_0 \delta_{k,1} \\ &\quad - \sum_{m=1}^{k-1} E_m \langle \varphi_0|S_0|\psi_{k-m}\rangle \\ &\quad - \sum_{m=0}^{k-1} E_m \langle \varphi_0|S_1|\psi_{k-m-1}\rangle), \quad k = 1, 2, \dots, \\ \kappa &= 1/\langle \varphi_0|S_0|\psi_0\rangle, \quad \langle \varphi_0| = \langle 0| + \langle 0|D_0 R, \end{aligned} \quad (2.11)$$

etc.

In practice, we may recommend a direct numerical use of Eqs. (2.10) and (2.11) as recurrences. Thus our knowledge of $|\psi_0\rangle$ [(2.6) and (2.8)] and $\langle \varphi_0|$ [(2.11)] enables us to define the first nontrivial contribution

$$E_1 = \kappa (\langle \varphi_0|W|\psi_0\rangle - h_0 - E_0 \langle \varphi_0|S_1|\psi_0\rangle) \quad (2.12)$$

to the energies as well as the subsequent wave function correction

$$|\psi_1\rangle = R(W|\psi_0\rangle - E_1 S_0|\psi_0\rangle - E_0 S_1|\psi_0\rangle). \quad (2.13)$$

For $k \geq 2$, we may proceed further in an iterative manner and, in a preparatory step, abbreviate

$$\begin{aligned} |\tilde{r}_1\rangle &= Q S_0|0\rangle, \quad |\tilde{r}_2\rangle = Q S_1|0\rangle, \\ \langle \tilde{r}_3| &= \langle \varphi_0|S_0 Q, \quad \langle \tilde{r}_4| = \langle \varphi_0|S_1 Q, \\ \langle \tilde{r}_5| &= \langle 0|W Q, \quad |\tilde{r}_a\rangle = E_0|\tilde{\psi}_1\rangle + E_1|\tilde{\psi}_0\rangle, \quad k = 2, \end{aligned} \quad (2.14)$$

where a restriction to the Q -projected subspace is marked by a tilde: this denotation is reasonable due to our normalization (2.8) (with $|\psi_k\rangle = |\tilde{\psi}_k\rangle$, $k \geq 1$, etc.)

In the forthcoming steps, our formulas (2.10) and (2.11) may be understood as an algorithm:

(A1) define

$$|\tilde{r}_b\rangle = \sum_{m=1}^{k-1} E_m |\tilde{\psi}_{k-m}\rangle; \quad (2.15)$$

(A2) compute

$$E_k = \kappa(\langle \tilde{r}_3 | \tilde{\psi}_{k-1} \rangle + \langle \tilde{\varphi}_0 | \tilde{W} | \tilde{\psi}_{k-1} \rangle - \langle \tilde{r}_3 | \tilde{r}_b \rangle - \langle \tilde{r}_4 | \tilde{r}_a \rangle); \quad (2.16)$$

(A3) redefine

$$|\tilde{r}_c\rangle = |\tilde{r}_b\rangle + E_k |\tilde{\psi}_0\rangle; \quad (2.17)$$

(A4) compute

$$|\tilde{\psi}_k\rangle = \tilde{R}(\tilde{W} |\tilde{\psi}_{k-1}\rangle - \tilde{S}_0 |\tilde{r}_c\rangle - \tilde{S}_1 |\tilde{r}_a\rangle - E_k |\tilde{r}_1\rangle - E_{k-1} |\tilde{r}_2\rangle); \quad (2.18)$$

(A5) redefine

$$|\tilde{r}_a\rangle := |\tilde{r}_c\rangle + E_0 |\tilde{\psi}_k\rangle, \quad k := k + 1; \quad (2.19)$$

and return to (A1) if the higher-order corrections are required.

In fact, our present GRS prescription is just an extension of the MRS formalism,² where $S = I$, i.e.,

$$|\tilde{r}_1\rangle = |\tilde{r}_2\rangle = |\tilde{r}_a\rangle = 0, \quad \langle \tilde{r}_3 | = \langle \tilde{\varphi}_0 |. \quad (2.20)$$

Similarly, we may even return to the standard RS $t = 0$ formalism with

$$\langle \tilde{r}_3 | = \langle \tilde{\varphi}_0 | = 0, \quad |\tilde{\psi}_0\rangle = 0, \quad |\tilde{r}_c\rangle = |\tilde{r}_b\rangle. \quad (2.21)$$

III. APPLICATION: THE PADÉ APPROXIMATION OF POTENTIALS

A phenomenological use of the various central potentials $V(r)$ in Eq. (1.3) ranges from nuclear physics up to molecular physics and quantum chemistry. The functions $V(r)$ may differ in their $r \rightarrow \infty$ asymptotics as well as in their (possibly even weakly singular) behavior at $r \rightarrow 0$. In general we may assume that an ansatz

$$V_{\text{Padé}}(r) = (\text{a polynomial in } z) / (\text{polynomial in } z), \quad z = z(r) = r^{\text{integer/integer}}, \quad (3.1)$$

$$\lim_{r \rightarrow 0} r^2 V_{\text{Padé}}(r) > -\frac{1}{4},$$

will represent a sufficiently broad class of the necessary approximants, making an arbitrary "realistic" input force expressible in the form

$$V(r) = V_{\text{Padé}}(r) + \lambda V_c(r), \quad (3.2)$$

where $\lambda V_c(r)$ is arbitrarily small.⁷

In Eq. (1.3) with $V = V_{\text{Padé}}$, a change of variables $r \rightarrow r^{\text{integer/integer}}$ may be used for its conversion into the same equation with the $z(r) = r^2$ "canonical" form of the Padé force with $\lambda V_c(r) = 0$,

$$V(r) = G(r^2) + A(r^2)/B(r^2), \quad A(x) = \sum_{m=0}^{p-1} a_m x^m, \quad G(x) = \sum_{m=1}^q g_m x^m, \quad g_q > 0, \quad B(x) = \sum_{m=0}^p b_m x^m > 0, \quad b_p > 0, \quad p \geq 0, \quad q \geq 1, \quad (3.3)$$

and with an inessential reinterpretation of the parameters: the angular momentum $l > -\frac{1}{2}$ becomes shifted and ceases to be an integer, and the original energy becomes transformed into one of the new couplings (say, g_{m_0}). Vice versa, the new energy variable E will represent just one of the original couplings. As an illustration, we may recall the screened Coulomb (SC) force

$$V_{\text{SC}}(r) = \alpha/r + \beta/(r + \gamma^2) \quad (3.4)$$

with the canonical representation (1.4).⁹

Among the merits of the Padé forces (3.3) we may mention not only their flexibility and universality, but also their elementary solvability for some particular couplings,¹⁰ useful for performing easily the numerical tests.¹¹ Nevertheless, their main merit lies in a possibility to use the harmonic oscillator basis $|n\rangle$ and analogy with Eq. (1.6). Thus our GRS approach to the $\lambda V_c = 0$ force (3.2) may be based on an arbitrary product decomposition of the denominator in (3.3),

$$B(x) = B_L(x) \cdot B_R(x),$$

$$B_i(x) = \sum_{m=0}^{p_i} b_m^{(i)} x^m, \quad p_i \geq 0, \quad i = L, R, \quad p_L + p_R = p, \quad (3.5)$$

followed by a multiplicative redefinition of our $\lambda V_c = W = 0$ equation of the type (1.5),

$$\left\{ B_L(r^2) \left[-\frac{d^2}{dr^2} + \frac{l(l+1)}{r^2} + G(r^2) - E \right] B_R(r^2) + A(r^2) \right\} \varphi(r) = 0, \quad (3.6)$$

$$\varphi(r) = \psi(r)/B_R(r^2),$$

and, finally, on an expansion of $\varphi(r)$ in the basis $|n\rangle$. As a result, we obtain our $W = 0$ chain-model generalization (2.1) of Eq. (1.6),

$$\sum_{n=m-t}^{m+t} \langle m | H | n \rangle \langle n | \varphi \rangle = \sum_{n=m-s}^{m+s} E \langle m | S | n \rangle \langle n | \varphi \rangle, \quad m = 0, 1, \dots, \quad (3.7)$$

where the operators

$$H = B_L(X) [H^{(\text{oscillator})} - X + G(X)] B_R(X) + A(X) = T, \quad S = B_L(X) B_R(X) = B(X) = S_0, \quad (3.8)$$

are band matrices since $X = r^2$ is tridiagonal. Obviously, we have $s = p$ and $t = p + q$ in general. For $q = 1$, we may even put $t = p$ after a rescaling of the coordinate such that $g_1 = 1$. Similarly, we may also incorporate $\lambda V_c \neq 0$.

In the conclusion, we have to reemphasize that our GRS series (1.10) need not represent the energy: For the $z(r) \neq r^2$ forces (3.1), $-E$ denotes, in fact, just a strength of the coupling g_0 that creates a bound state at a prescribed energy $-g_{m_0}$, $m_0 > 0$. Hence we are forced to denote $G_0(r) = G(r) - g_{m_0} r^{2m_0} + g_0$, $E = -g_{m_0}$ and change the corresponding matrices in Eq. (3.7),

$$H = B_L(X) [H^{(\text{oscillator})} - X + G_0(X)] B_R(X) + A(X) = T, \quad (3.9)$$

$$S = B_L(X) X^{m_0} B_L(X) = X^{m_0} B(X) = S_0.$$

Of course, we have to redefine also the integer $s = p + m_0$ and add the corrections $\lambda V_c(r) \neq 0$ and $\lambda S_1 \neq 0$ if needed.

IV. THE CONSTRUCTION OF PROPAGATORS R IN TERMS OF THE MATRIX CONTINUED FRACTIONS

A. The formalism

In accord with Sec. II, a knowledge of R (2.5) not only leads to the zeroth-order solution of the unperturbed problem, but it also renders possible an explicit specification of all the higher-order corrections. Thus a practical efficiency of the GRS expansions will mainly be determined by the inversion in (2.5).

Keeping this in mind, let us partition the unperturbed Schrödinger operator into its $(t \times t)$ -dimensional submatrices,

$$D_0 = T - E_0 S_0 = \begin{pmatrix} a_0 & b_0 & & \\ c_1 & a_1 & b_1 & \\ & c_2 & a_2 & b_2 \\ & & \dots & \end{pmatrix}, \quad (4.1)$$

and also introduce the partitioned auxiliary matrices,

$$F_U = \begin{pmatrix} I & b_0 f_1 & & \\ & I & b_1 f_2 & \\ & & I & b_2 f_3 \\ & & & \dots \end{pmatrix},$$

$$F_L = \begin{pmatrix} I & & & \\ f_1 c_1 & I & & \\ & f_2 c_2 & I & \\ & \dots & & \end{pmatrix}, \quad (4.2)$$

$$F_D = \begin{pmatrix} 1/f_0 & & & \\ & 1/f_1 & & \\ & & \dots & \end{pmatrix}.$$

Then we may postulate an explicit factorization formula

$$D_0 = T - E_0 S_0 = F_U F_D F_L. \quad (4.3)$$

This becomes an algebraic identity if and only if

$$1/f_k = a_k - b_k f_{k+1} c_{k+1}, \quad k = 0, 1, \dots \quad (4.4)$$

Thus, whenever D_0 may be treated as an infinite-dimensional limit of its truncated forms, the latter relations may be initialized by $f_{M+1} = 0$, $M \rightarrow \infty$ and define just a matrix form of the analytic continued fractions.¹²

A motivation of our matrix continued fractional (MCF) factorization (4.3) of D_0 lies in the related simple form of the inverse matrices (e.g.,

$$F_U^{-1} = \begin{pmatrix} I & -b_0 f_1 & b_0 f_1 b_1 f_2 & -b_0 f_1 b_1 f_2 b_2 f_3 \dots \\ & I & -b_1 f_2 & -b_1 f_2 b_2 f_3 \dots \\ & & I & -b_2 f_3 \dots \\ & & & \dots \end{pmatrix}, \quad (4.5)$$

etc.) and also in the easy projection of the triangular matrices

(e.g., $F_L Q = Q F_L Q$, etc.). Both these consequences of our ansatz (4.2) lead quickly to the explicit MCF definition

$$R = F_L^{-1} Q \begin{pmatrix} f_0 & & \\ & f_1 & \\ & & f_2 \\ & & & \dots \end{pmatrix} Q F_U^{-1} \quad (4.6)$$

of our "input" unperturbed propagator R .

B. The MCF convergence

For the systems of the simple type (3.9), an important merit lies in a possibility of their semi- or non-numerical analysis. In particular, a study of convergence of the present infinite MCF expansions

$$f_k = \frac{1}{a_k - b_k \frac{1}{a_{k+1} - \dots} c_{k+1}} \quad (4.7)$$

becomes feasible¹³ due to a validity of the same leading-order asymptotic behavior of the matrices H and S ,

$$\langle M + m | D_0 | M + n \rangle = \text{const } M^{p+q} \binom{2t}{t+m-n} + O(M^{p+q-1}), \quad M \gg 1. \quad (4.8)$$

This may quite easily be derived from the well known formula^{2,5}

$$\langle m | r^2 | n \rangle = \begin{cases} 2m + l + \frac{3}{2}, & n = m, \\ [(m+1)(m+l+\frac{3}{2})]^{1/2}, & n = m+1, \\ [m(m+l+\frac{1}{2})]^{1/2}, & n = m-1, \end{cases} \quad (4.9)$$

and, up to the irrelevant constant term, leads to the leading-order MCF estimate

$$f_{M+k} \sim f_M^{(\text{FPO})} \sim M^{-p-q} \frac{1}{a^{(0)} - b^{(0)} \frac{1}{a^{(0)} \dots} c^{(0)}},$$

$$a^{(0)} = S S^T + S^T S, \quad b^{(0)} = c^{(0)T} = S^T S^T, \quad (4.10)$$

$$S_{mn} = \binom{t}{n-m}, \quad m, n = 1, 2, \dots, t,$$

which may be summed up in the closed form

$$f_M^{(\text{FPO})} \sim M^{-p-q} (S S^T)^{-1}. \quad (4.11)$$

In a higher-order analysis, we may recall the specialized literature¹⁴ and create the more general asymptotics [the so-called fixed-point MCF expansions] of the type

$$f_{M+k} = f_M^{(\text{FPO})} + f_M^{(\text{FP1})} + \dots + f_M^{(\text{FPk})} + f'_{M+k}. \quad (4.12)$$

Here the remainder term is a MCF-type quantity again: for a sufficient number k_{FP} of subtractions of the type (4.12), the corresponding remainder recurrences may be linearized for $M \gg 1$ and the rigorous proof of convergence becomes trivial.¹³⁻¹⁵

In the present context, let us illustrate the higher-order MCF analysis on a particular $q > 2$ and $\gamma > 0$ extension of the asymptotic estimate (4.8),

$$\begin{aligned} & \text{const} \langle M+m | D_0 | M+n \rangle \\ &= M^{p+q} \binom{2t}{t+m-n} \\ &+ \gamma M^{p+q-1} \binom{2t-2}{t+m-n-1} + O(M^{p+q-2}), \\ &\gamma = b_{p-1}/b_p + g_{q-1}/g_q, \quad M \gg 1. \end{aligned} \quad (4.13)$$

Then, in accord with the Appendix of Ref. 13, we may postulate

$$\begin{aligned} f_{M_0-k} &= \frac{\text{const } M_0^{-p-q}}{SS^T + S^T |\beta\rangle \rho_k \langle \beta| S + \text{small terms}}, \\ \langle \beta | &= (1, -1, 1, -1, \dots, (-1)^{t+1}), \\ M_0 &\gg 1, \quad k = 0, 1, \dots \end{aligned} \quad (4.14)$$

Such an approximation dominates the second-order behavior and reduces the $(t \times t)$ -dimensional MCF recurrences to mere one-dimensional mapping $\rho_k \rightarrow \rho_{k+1}$, $k \ll M_0$,

$$\begin{aligned} \rho_{k+1} &= (\rho_k + \gamma M_0^{-1} + O(M_0^{-2})) / (1 + t\rho_k), \\ k &= 0, 1, \dots \end{aligned} \quad (4.15)$$

An analysis of convergence of the latter reduced MCF mapping is not difficult. In accord with Fig. 1, we may conclude that it has a pair of the real fixed points,

$$\rho_k \approx \rho_{k+1} \approx \pm \rho^{(\text{FP})} = \pm [\gamma/M_0 t + O(M_0^{-2})]^{1/2}, \quad k \gg 1, \quad (4.16)$$

the positive one of which is a point of accumulation of the sequence

$$\rho_0 \rightarrow \dots \rightarrow \rho_k \rightarrow \rho_{k+1} \rightarrow \dots, \quad k \ll M_0, \quad (4.17)$$

for an almost arbitrary initial choice ρ_0 . In particular, a convergence from above and from below may be expected for $|\rho_0| > \rho^{(\text{FP})}$ and $|\rho_0| < \rho^{(\text{FP})}$, respectively.

V. THE NON-NUMERICAL CONSTRUCTION OF $R = \hat{R}$

A. The second auxiliary rearrangement operator $\hat{\Theta}$ and the final form of the GRS formalism

For the sufficiently large number of terms in the fixed point formula (4.12), we may define

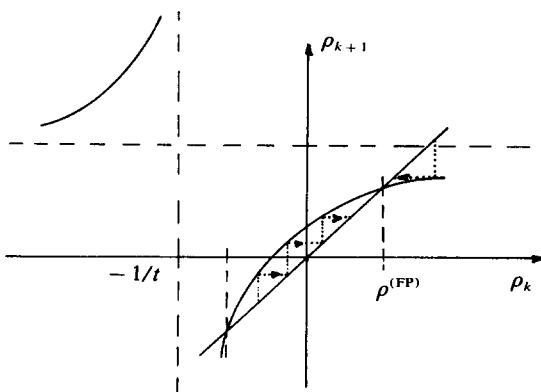


FIG. 1. The auxiliary $(t \times t)$ -dimensional continued fraction convergence in the leading-order asymptotic one-dimensional approximation (4.15) $[\rho^{(\text{FP})} = (\gamma/M_0 t)^{1/2}]$.

$$\hat{f}_M = f_M^{(\text{FP}0)} + f_M^{(\text{FP}1)} + \dots + f_M^{(\text{FP}k)}, \quad M \gg M_0, \quad (5.1)$$

as a sufficiently reliable approximant to the uncapped MCF quantities. Vice versa, the capped quantities $\hat{f}_M$ have to correspond to some unknown operator $\hat{D}_0$ via the factorization (4.3). Now, an assumption $\hat{b}_m = b_m$, $\hat{c}_{m+1} = c_{m+1}$ may be combined with the necessary and sufficient condition (4.4) into an explicit definition

$$\hat{a}_m = \hat{f}_m^{-1} + b_m \hat{f}_{m+1} c_{m+1}, \quad m \gg M_0, \quad (5.2)$$

of the missing matrix elements of $\hat{D}_0$.

For the sufficiently large M_0 or k_{FP} [= number of terms in (5.1)], the difference between the capped and uncapped operators will be small,

$$\hat{\Theta} = \hat{D}_0 - D_0 = O(\lambda), \quad (5.3)$$

and may be treated as another component of the perturbation. In more detail, we may replace Eq. (1.9) by Eq. (1.11), i.e., (a) start from the FP formula (5.1) and define the capped operator $\hat{D}_0$ by means of Eq. (5.2); (b) define the capped, solvable band matrix $\hat{T} = \hat{T}(\hat{f}_i, i = 0, 1, \dots)$,

$$\hat{T} = T + \hat{D}_0 - D_0, \quad (5.4)$$

and preserve the overlap matrix S unchanged; (c) use Eq. (4.6) and replace R [the inversion (2.5)] by the capped operator $\hat{R} = \hat{R}(\hat{f}_i, i = 0, 1, \dots)$; (d) replace all D_0 by $\hat{D}_0$ and also substitute

$$\lambda \hat{W} = H - \hat{T} = \lambda W - \hat{\Theta}, \quad \hat{\Theta} = O(\lambda), \quad (5.5)$$

for all the uncapped λW 's; and (e) use the GRS formalism of Sec. II with the capped quantities: $T \rightarrow \hat{T}$, $R \rightarrow \hat{R}$, $D_0 \rightarrow \hat{D}_0$, and $W \rightarrow \hat{W}$.

B. The Padé oscillator illustration

Let us put $V_c(r) = 0$ in Eq. (3.2) and define the related uncapped unperturbed operator R in terms of the MCF quantities f_n . Then, our perturbation $\lambda H_1 = -U \neq 0$ is separable and reflects merely our "bad" choice of E_0 . Of course, we are tempted to define now the "best" value of $E_0 = E_{\text{exact}}$ by the requirement $g(E_0) = 0$, i.e., $\lambda H_1 = 0$, i.e.,

$$\det 1/f_0 = 0 \quad (5.6)$$

[cf. Eq. (2.7)]. This is a Brillouin-Wigner-type formula and enables us to determine E_{exact} numerically by the trial and error technique (cf., e.g., Graffi and Grecchi,¹⁶ etc.).

In light of Sec. IV B, acceleration of the MCF convergence in Eq. (5.6) may be achieved by the FP subtractions. Alternatively, we may ignore the higher-order MCF corrections and use Eq. (5.1) in an approximative evaluation of the energies. In the present GRS context, we may combine both these ideas: We may start from the uncapped $W = 0$ and matrix D_0 and redefine the corrections ($\lambda \hat{W} \neq 0$),

$$\begin{aligned} & \text{const} \langle M_0 + m | \lambda \hat{W} | M_0 + n \rangle \\ &= \gamma M_0^{p+q-1} \binom{2t-2}{t+m-n-1} + O(M_0^{p+q-2}), \end{aligned}$$

$$m, n \geq 0,$$

via the explicit algebraic formula (5.1). No infinite MCF expansions of the type (4.7) are needed anymore.

C. Perturbative determination of the upper and lower estimates of energies

An introduction of the auxiliary operator $\hat{\Theta} = O(\lambda)$ and of the secondary rearrangement (1.11) of the full Hamiltonian H has been motivated formally: It enables us to reconstruct the complete GRS input (namely, the operator $R = \hat{R}$, approximant $\hat{T}$, and the split matrices $H_0 = \hat{T} + U$ and $\lambda H_1 = \lambda \hat{W} - U$) in an entirely non-numerical manner.

Now, let us pay attention to the corresponding GRS perturbation formulas at a given and fixed order of precision $O(\lambda^k)$. For the sake of simplicity, we shall analyze just the first-order example of the preceding subsection and notice that the leading-order components (5.7) of the perturbation $\lambda \hat{W}$ are positive semidefinite. Indeed, they may be interpreted as certain matrix elements of the powers of $r^2 > 0$.

The most important consequence of the above observation lies in an easy majorization and minorization of the whole perturbation $\lambda \hat{W} > 0$ by the operators

$$\begin{aligned} \lambda \hat{W}^{(\text{maj})} &= \lambda z^{(\text{maj})} \hat{W}, \quad z^{(\text{maj})} > 1 + O(1/M_0), \\ z^{(\text{min})} &< 1 - O(1/M_0). \end{aligned} \quad (5.8)$$

The corresponding replacement of the parameter $\gamma > 0$ in Eq. (5.7) by $\gamma^{(\text{maj})} = z^{(\text{maj})}\gamma > 0$ and $\gamma^{(\text{min})} = z^{(\text{min})}\gamma > 0$, respectively, leads also to an explicit change of ρ_0 in (4.14),

$$\begin{aligned} \rho_0^{(\text{maj})} &\in (\rho_a^{(\text{maj})}, \rho_b^{(\text{maj})}), \\ \rho_a^{(\text{maj})} &= (z^{(\text{maj})}\gamma/M_0 t - |\text{const}|/M_0^2)^{1/2}, \\ \rho_b^{(\text{maj})} &= (z^{(\text{min})}\gamma/M_0 t + |\text{const}|/M_0^2)^{1/2}. \end{aligned} \quad (5.9)$$

Here, the length of the uncertainty interval need not be negligible since the parameters themselves are small.

For the purely numerical purposes, let us now treat ρ_0 as a free parameter, and eliminate also the GRS convergence questions by using the Brillouin–Wigner-type formula (5.6). We may expect that the resulting energy roots $E = E(M_0, \rho_0)$ will also follow the majorization and minorization change of the Hamiltonians,

$$E(M_0, \rho_0^{(\text{min})}) < E_{\text{exact}} < E(M_0, \rho_0^{(\text{maj})}). \quad (5.10)$$

As a consequence, the best results will lie somewhere within the interval of parameters

$$\rho_0 \in (\rho_b^{(\text{min})}, \rho_a^{(\text{maj})}). \quad (5.11)$$

This is our first important observation.¹⁷

In the computations, a precise localization of the optimal interval (5.11) may prove difficult both algebraically (the higher-order FP corrections are complicated¹⁴) and computationally (whenever we get $\rho_b^{(\text{min})} \approx \rho_a^{(\text{maj})}$ for some small M_0). Nevertheless, there is an easy way out of this difficulty: The dependence of energies $E(M_0, \rho_0)$ weakens with the increasing M_0 so that the two curves $E(M_0^{(\alpha)}, \rho_0)$ and $E(M_0^{(\beta)}, \rho_0)$ must intersect somewhere in the interval (5.11). The point of intersection specifies the best values of ρ_0 and E_0 geometrically.

Numerically, the geometric “sandwiching” of energies of this type has been tested on the various $t \leq 2$ examples with $M_0^{(\alpha)} = 400$ and $M_0^{(\beta)} = 500$. Schematically, the results are summarized in Fig. 2 where a common pattern of the M_0 and

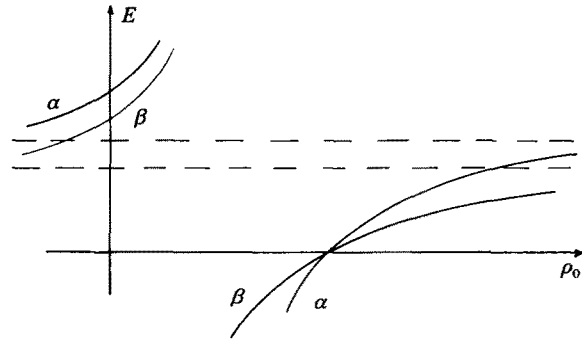


FIG. 2. A geometric, numerical sandwiching of the energies.

ρ_0 dependence of $E(M_0, \rho_0)$ is depicted in arbitrary units. The scheme worked even beyond its natural limits given by its present derivation: A large initial choice of $\rho_0 \rightarrow \pm \infty$ simulates the MCF initialization $f_{m_0} = 0$ and exhibits also the expected variational behavior typical for the standard increasing truncation of Hamiltonians. For the smaller M_0 's, a use of the third curve $E(M_0^{(\gamma)}, \rho_0)$ may sometimes be recommended for an estimate of the error bars.¹⁸

VI. SUMMARY

In the paper, a generalized Rayleigh–Schrödinger perturbation theory has been proposed and described as a means of solving the Schrödinger bound-state problem when formulated as a matrix equation in some nonorthogonal basis,

$$H\psi = E S \psi, \quad S \neq I. \quad (6.1)$$

As an illustrative example, we have chosen the determination of bound states in a general central potential approximated by a ratio of two polynomials of an arbitrary degree. The specific features of such a Padé oscillator example proved extremely useful.

(1) In the first stage, a band-matrix $(2t + 1)$ -diagonal structure of the related Schrödinger operator enabled us to satisfy the main and strongest GRS requirement (namely, an *a priori* knowledge of the unperturbed propagator R) in the manner proposed recently for the $S = I$ case [namely, via a construction of R in terms of certain auxiliary $(t \times t)$ -dimensional continued fractions].

(2) An asymptotic smoothness of the matrix elements of D_0 proved essential for a subsequent elimination of the infinite (i.e., numerical) MCF expansions. In the GRS formulas, we have replaced them by certain algebraic (so-called fixed point, capped) finite expressions that generalized directly the standard RS input unperturbed spectrum.

(3) The $S \neq I$ formulation of the Padé-oscillator examples with the FP elimination of the MCF expansions has been shown to contain the positive semidefinite perturbations $\lambda \hat{W}$. The subsequent easy majorization and minorization (sandwiching) of Hamiltonians also leads quickly to the GRS formulas that majorize and minorize the energies of a $O(\lambda^k)$ level of precision in principle.

(4) A modified interpretation of the operator S renders possible a GRS generation of the perturbation formulas for the couplings

$$G = G_0 + \lambda G_1 + \lambda^2 G_2 + \cdots \quad (6.2)$$

Such a completion of the expansions of energies may also be complemented by the perturbative construction of the so-called “Sturmians”⁵ if needed.

In a broader methodical sense, all our Padé-oscillator results exemplify the main merits of our GRS expansions: (1) a feasibility of the non-numerical $S \neq I$ expansions, (2) a generalization and weakening of the standard RS “solvability” requirement, (3) a possible sandwiching of the exact values in a variational and “antivariational” manner, and (4) an extension (6.2) of the standard “changing-energy” picture of vicinity of the fixed “unperturbed” system to a full analysis [cf. Eq. (3.9) with $m_0 = 0, 1, \dots, q$] of mutual relations between the variable couplings and the energies.

¹E. Merzbacher, *Quantum Mechanics* (Wiley, New York, 1961).

²M. Znojil, Phys. Rev. A **35**, 2448 (1987); Phys. Lett. A **125**, 443 (1987); preprint JINR Dubna E5-87-634, 1987; M. Znojil, M. Flynn, and R. F. Bishop, proceedings “Hadron Structure 87,” in *Physics and Applications*, edited by D. Krupa (Institute of Physics, Bratislava, 1988), Vol. 14, p. 252.

³See, e.g., the references listed by Y. P. Varshni, Phys. Rev. A **36**, 3009 (1987).

⁴N. Bessis and G. Bessis, J. Math. Phys. **21**, 2780 (1980).

⁵R. R. Whitehead, A. Watt, G. Flessas, and M. A. Nagarajan, J. Phys. A: Math. Gen. **15**, 1217 (1982).

⁶H. Wilkinson, *The Algebraic Eigenvalue Problem* (Oxford U.P., Oxford, 1965).

⁷G. A. Baker, Jr., and P. Graves-Morris, *Padé Approximants* (Addison-Wesley, Reading, MA, 1981).

⁸M. Znojil, J. Math. Phys. **25**, 2979 (1984).

⁹M. Znojil, Phys. Lett. A **94**, 120 (1983).

¹⁰M. Znojil, J. Phys. A: Math. Gen. **16**, 279 (1983).

¹¹M. H. Blecher and P. G. L. Leach, J. Phys. A: Math. Gen. **20**, 5923 (1987).

¹²H. S. Wall, *Analytic Theory of the Continued Fractions* (Van Nostrand, New York, 1948).

¹³M. Znojil, J. Phys. A: Math. Gen. **17**, 1611 (1984).

¹⁴See Ref. 8 and further papers listed in M. Znojil, Czech. J. Phys. B **37**, 1072 (1987).

¹⁵M. Znojil, Phys. Rev. D **24**, 903 (1981).

¹⁶S. Graffi and V. Grecchi, Lett. Nuovo Cimento **12**, 425 (1975).

¹⁷M. Znojil, preprint JINR Dubna E4-86-60, 1986.

¹⁸In fact, there exist many other majorizations and minorizations of energies in perturbation theory, cf., e.g., R. C. Young, L. C. Biedenharn, and E. Feenberg, Phys. Rev. **106**, 1151 (1957), as one of the typical early references also using continued fractions in a somewhat different setting.